

CS443 - Network Security Assignment 6: Network Telescope Analysis

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This report shows the results of analyzing network traffic from a telescope sensor. The goal is to find scanning behavior, top scanner IPs, and protocol use. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 presents the ten most active scanning sources observed in the dataset. The top scanners were identified based on the volume of TCP SYN and UDP packets they generated. These ten IPs alone accounted for over 9% of total scanning traffic. The highest contributor, 185.242.226.50, sent more than 188,000 packets, indicating likely use of automated scanning tools or botnet infrastructure.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Target Port Analysis

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (e.g., 22, 80, 443), while UDP scanning targeted DNS, SIP, and network service discovery. These results reflect broad automated probing for known services.

Port	Protocol	Packets
22	TCP	174,938
3389	TCP	101,839
80	TCP	90,764
8088	TCP	82,746
8080	TCP	80,344
8728	TCP	76,159
5555	TCP	69,330
443	TCP	53,354
8122	TCP	51,620
12103	TCP	50,686
514	TCP	144,733
514	UDP	144,733
53	UDP	91,129
5060	UDP	63,392
1194	UDP	40,973
32410	UDP	27,957
123	UDP	23,558
55001	UDP	13,926
1900	UDP	13,211
6881	UDP	13,009
5353	UDP	11,856

Table 2: Top 10 destination ports for TCP and UDP scans

2.1 TCP Port Targets

Table 3 shows the most targeted TCP ports. Port 8124 dominates with over 7,000 packets, likely targeting specific IoT devices or web services running on non-standard ports.

Port	TCP Packets
8124	7,150
6066	2,966
8000	2,710
34567	2,538
37777	2,536
8088	1,681
84	1,402
22	1,202
12102	1,135
3389	1,074

Table 3: Most targeted TCP destination ports

Notable observations include:

- Port 22 (SSH) - Standard administrative access
- Port 3389 (RDP) - Windows Remote Desktop
- Ports 8000, 8088, 8124 - Alternative HTTP/web services

- High-numbered ports (34567, 37777) - Likely IoT or P2P applications

2.2 UDP Port Targets

Table 4 reveals UDP scanning patterns focused on network services and protocols commonly used for amplification attacks.

Port	UDP Packets
5060	3,878
514	1,522
53	911
123	386
10184	291
8000	185
8082	148
5353	142
6881	138
4000	137

Table 4: Most targeted UDP destination ports

The UDP port distribution reveals targeting of:

- Port 5060 (SIP) - VoIP communication protocol
- Port 514 (Syslog) - System logging service
- Port 53 (DNS) - Domain name resolution
- Port 123 (NTP) - Network time protocol
- Port 5353 (mDNS) - Multicast DNS

These ports are commonly exploited for DDoS amplification attacks due to their potential for generating large response packets.

3 Protocol Distribution

The protocol breakdown analysis would show the relative distribution of TCP versus UDP traffic, providing insights into the types of scanning activities predominant in the dataset.

4 Heavy Hitter Analysis

The most prolific scanner, 88.213.209.194, originates from Bulgaria and demonstrates characteristics typical of automated reconnaissance tools. This source generated 7,598 packets targeting 832 unique IP addresses across only three destination ports, indicating a focused scanning strategy.

The limited port range suggests this scanner employs a targeted approach rather than comprehensive port sweeping, possibly searching for specific vulnerable services or devices. The high IP address diversity (832 unique targets) combined with the concentrated port selection indicates an efficient scanning methodology designed to maximize coverage while minimizing detection.

5 Traffic Visualizations

Figure 1: Temporal distribution of scanning activity using 5-minute time bins

Figure 2: Distribution analysis of the most frequently scanned ports

Figure 3: Target diversity analysis for the most active scanning sources

6 Security Analysis and Interpretation

The observed scanning patterns reveal several concerning trends in internet-wide reconnaissance activities:

Automated Mass Scanning: The concentration of traffic from a small number of sources, combined with the consistent targeting of specific ports, strongly indicates automated scanning tools rather than manual reconnaissance. This automation enables attackers to efficiently survey large address spaces for vulnerable services.

Service-Specific Targeting: The port distribution shows attackers focusing on specific services including SSH (22), RDP (3389), SIP (5060), and various web services on non-standard ports. This targeted approach suggests attackers are searching for known vulnerable implementations or default configurations.

IoT and Infrastructure Focus: Many targeted ports (8124, 34567, 37777) correspond to IoT devices, IP cameras, and embedded systems that often have poor security practices. The prevalence of these targets indicates ongoing campaigns to build IoT botnets.

DDoS Preparation: The significant UDP scanning activity, particularly targeting DNS (53), NTP (123), and SIP (5060) services, suggests reconnaissance for potential DDoS amplification attacks. These protocols can generate large response packets, making them valuable for reflection attacks.

Persistence and Scale: The steady traffic pattern over time, as shown in [Figure 1](#), indicates continuous scanning operations rather than sporadic attacks. This persistence suggests well-resourced threat actors operating scanning infrastructure.

7 Threat Assessment

While network telescopes themselves face no direct risk from this traffic (being non-routable), the observed patterns indicate active threat reconnaissance that poses risks to legitimate internet infrastructure:

- **High Risk:** The scale and persistence of scanning suggests professional threat actors
- **Botnet Activity:** Heavy hitters like 88.213.209.194 likely represent compromised hosts in larger botnets

- **Service Targeting:** Focus on specific ports indicates knowledge of vulnerable services and exploitation techniques
- **Amplification Potential:** UDP port targeting reveals preparation for large-scale DDoS attacks

Organizations should monitor for similar scanning patterns against their networks and ensure proper security controls are in place for the identified high-risk services.